

M4H1. Definition de Architecture

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Project Planning

1. Objective

Measure noise pollution on three floors of a college building and log all readings into a single time-stamped CSV.

2. System Architecture

- Three ESP32 nodes, one per floor, each with a KY-038 analog sound sensor.
- Nodes send their readings over ESP-NOW to a fourth ESP32 that acts as a gateway.
- The gateway connects to the Pi over USB and forwards each packet as a serial line.
- The Pi parses the serial stream and appends rows to the CSV.
- A gateway is needed because ESP-NOW works only between ESP devices; the Pi cannot receive those frames.

3. Hardware per Node

- ESP32 development board (ESP32-WROOM-32 or equivalent), USB powered.
- KY-038 sound sensor, analog output A0 to GPIO34 (ADC1_CH6), powered from 3.3 V.
- Sensitivity potentiometer tuned per node during calibration.

4. Data Format

CSV columns: Timestamp, Node_ID, Info_Sens_1, Info_Sens_2, Info_Sens_3, Label.

- Timestamp: assigned by the Pi on reception, ISO 8601 with milliseconds.
- Node_ID: 1, 2 or 3, hard-coded in each node's firmware.
- Info_Sens_1 to Info_Sens_3: three raw ADC readings (0-4095) from the same sampling window.
- Label: floor identifier (floor_1, floor_2, floor_3), derived from Node_ID by the logger.

5. Firmware Plan

- Nodes: register the gateway MAC as an ESP-NOW peer at startup.
- Nodes: once per second, take three ADC readings and send them with the Node_ID in a fixed struct.
- Nodes: check the ESP-NOW delivery callback and retry on failure.
- Gateway: on each received packet, print one fixed-format line to serial at 115200 baud.

6. Raspberry Pi Logger

- Python service that reads the serial port, validates each line, and appends to the CSV.
- Writes the header if the file is new and flushes every row to survive power loss.
- Runs as a systemd service so logging restarts after a reboot.

7. Testing and Deployment

- Bench test all three nodes next to the gateway before installing anything.
- Verify ESP-NOW range from each floor to the chosen gateway location.
- Install one node per floor, record which Node_ID is where, and run a 24-hour capture.

8. Risks

- Range through concrete floors may be poor; mitigate by placing the gateway in a stairwell.
- The KY-038 is uncalibrated, so results are comparative between floors, not absolute decibels.
- Simultaneous transmissions may overrun the serial buffer; stagger the sampling offset per node.